



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 16/10/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 10

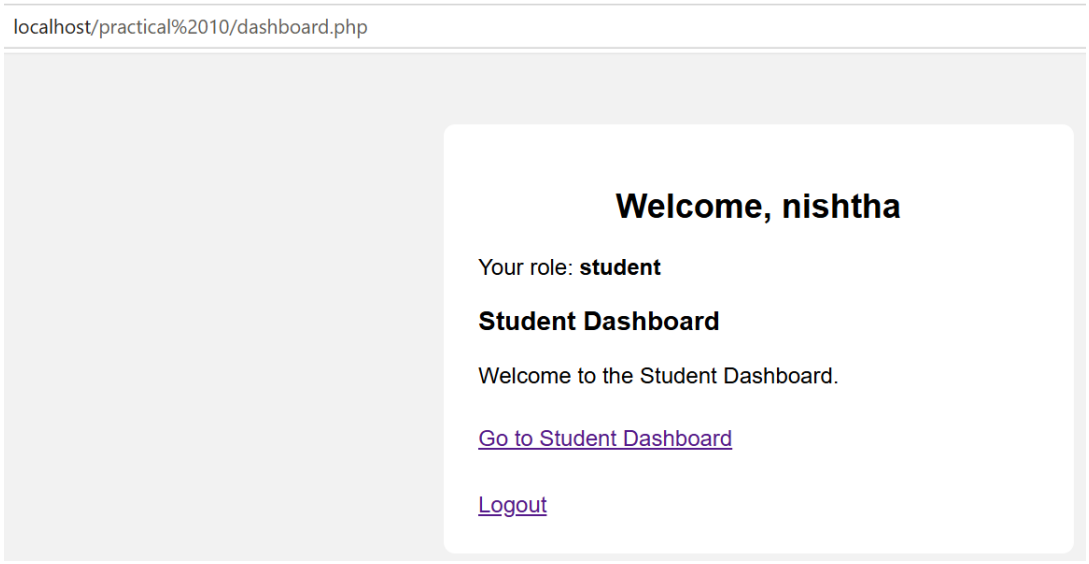
Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token

Problem Statement: - Develop secure login and logout functionality using PHP sessions. Verify password using password_verify(), regenerate session ID after login, restrict dashboard access, implement role-based redirection, session timeout, and optional secure remember-me token.

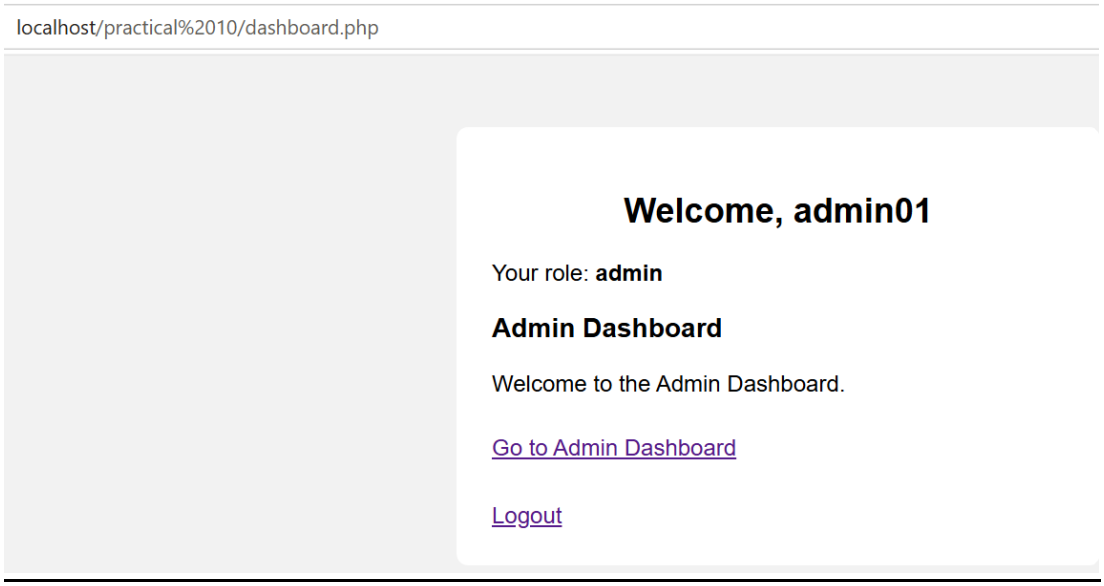
Objective :- To develop a secure login and logout system using PHP sessions by verifying user passwords with password_verify(), regenerating session IDs after successful login, restricting unauthorized access to the dashboard, implementing role-based redirection and session timeout, and optionally providing a secure Remember Me feature using tokens.

Screenshot

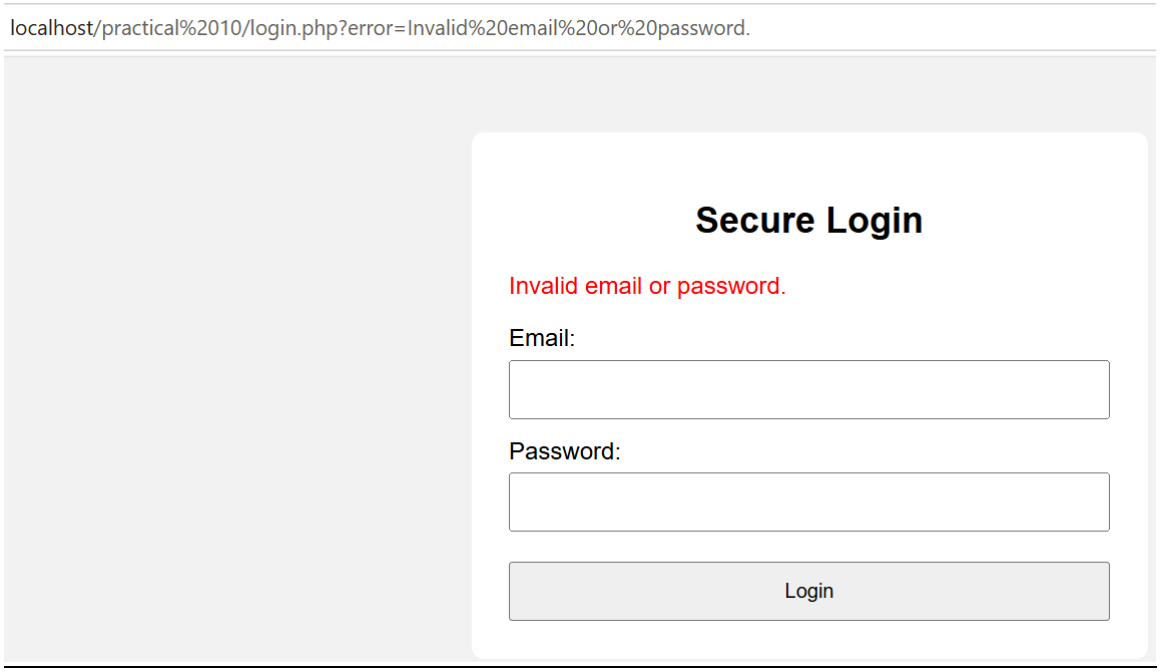
Correct student login:



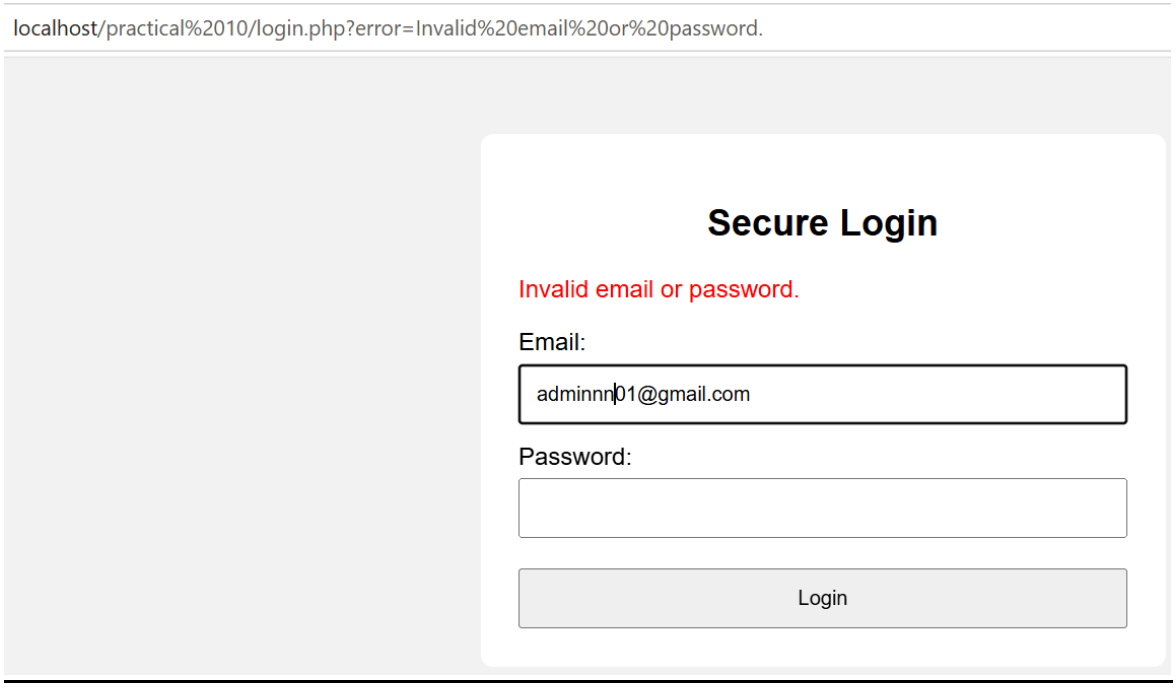
Correct admin login:

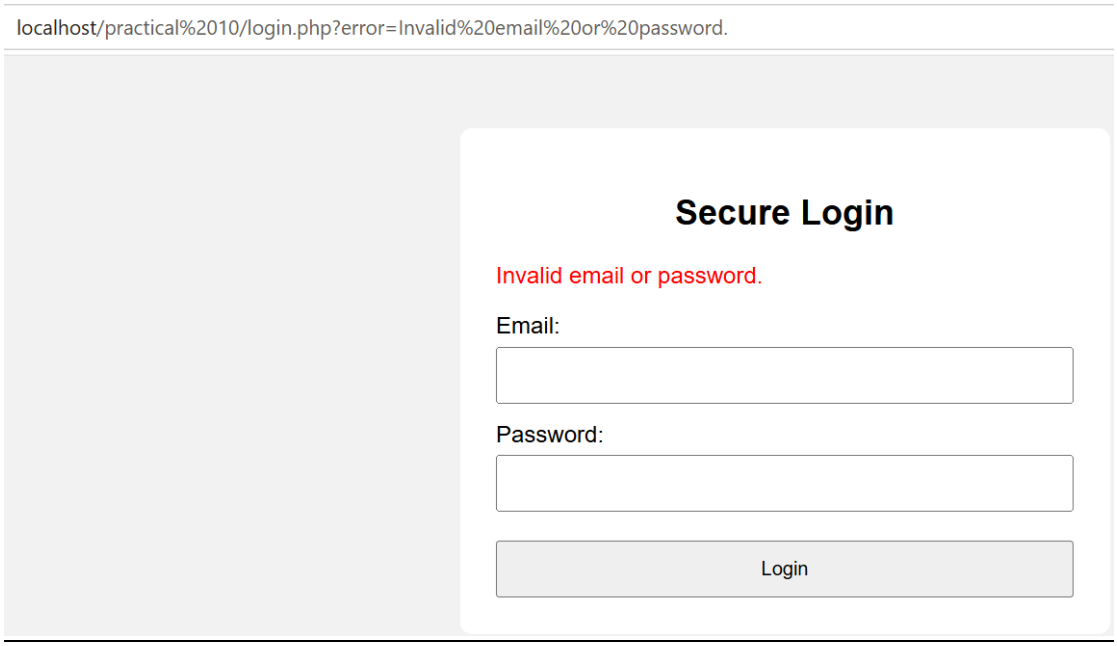


Wrong password :

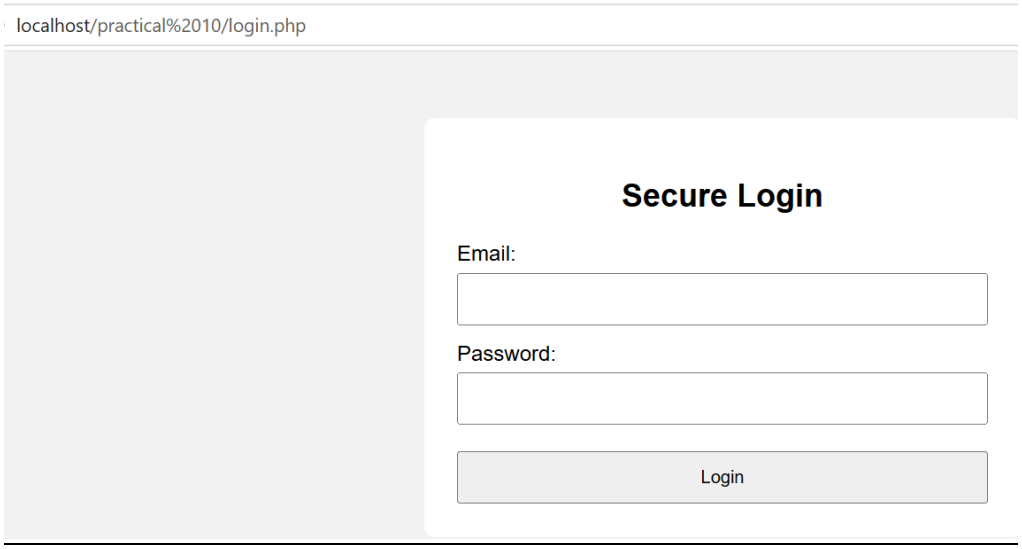


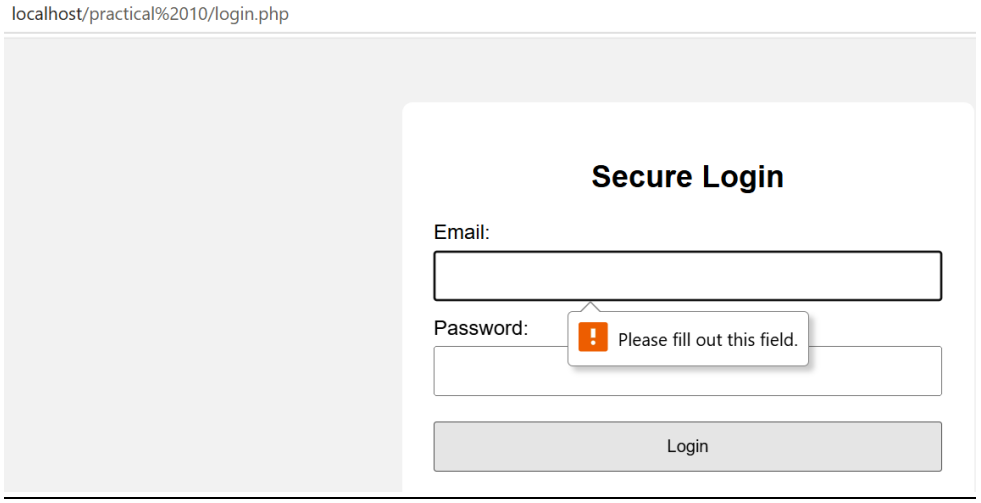
invalid email validation:




localhost/practical%2010/login.php?error=Invalid%20email%20or%20password.
Secure Login
Invalid email or password.
Email:
Password:
Login

Empty field :


localhost/practical%2010/login.php
Secure Login
Email:
Password:
Login


localhost/practical%2010/login.php
Secure Login
Email:
Password:
Please fill out this field.
Login

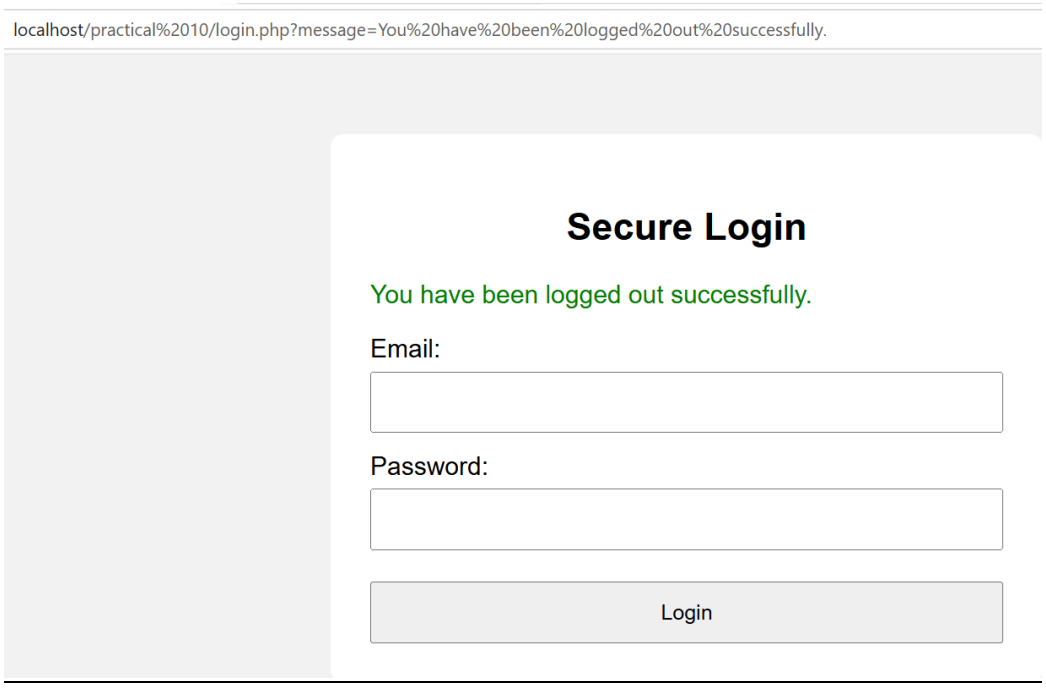
Student opens admin dashboard:



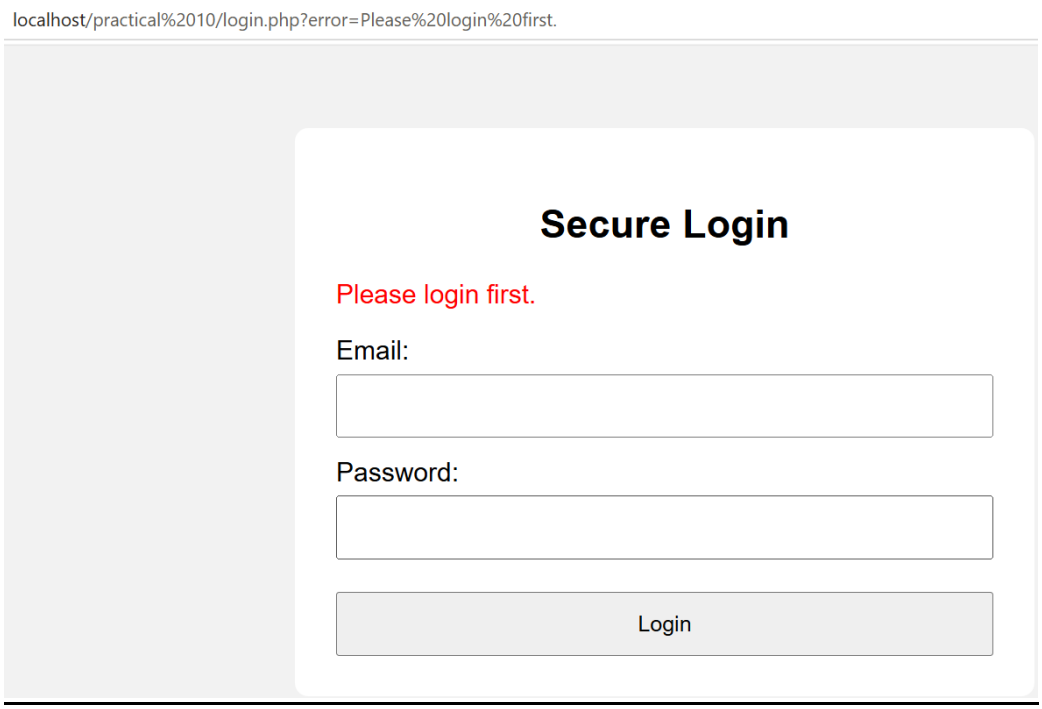
Admin opens student dashboard:



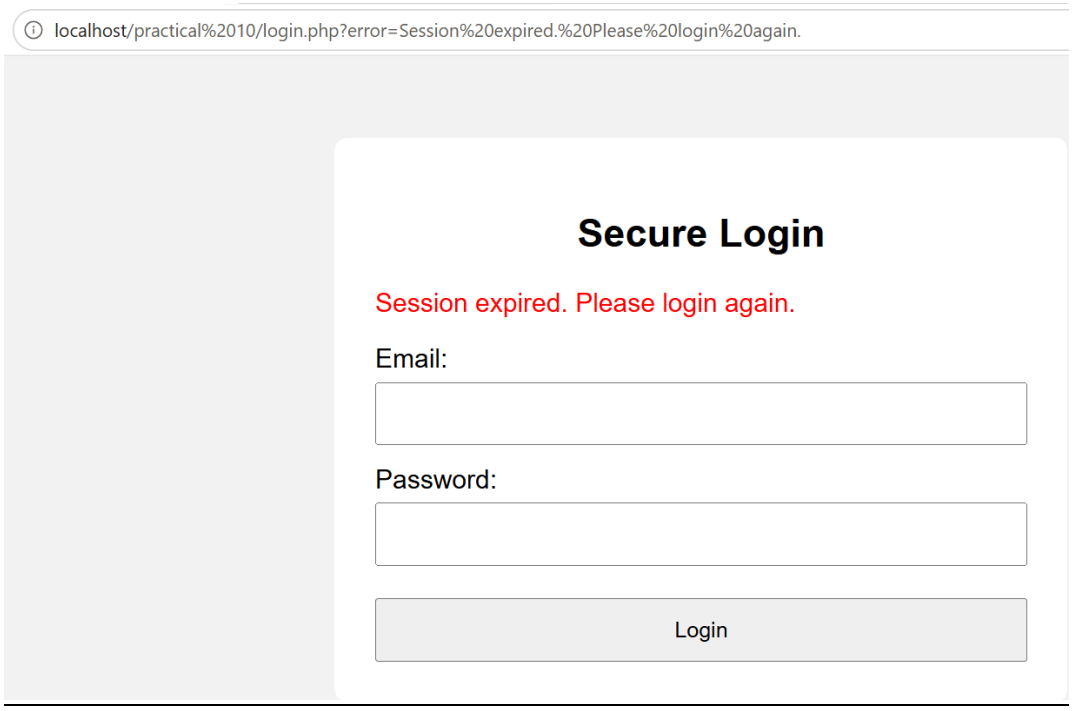
Logout:



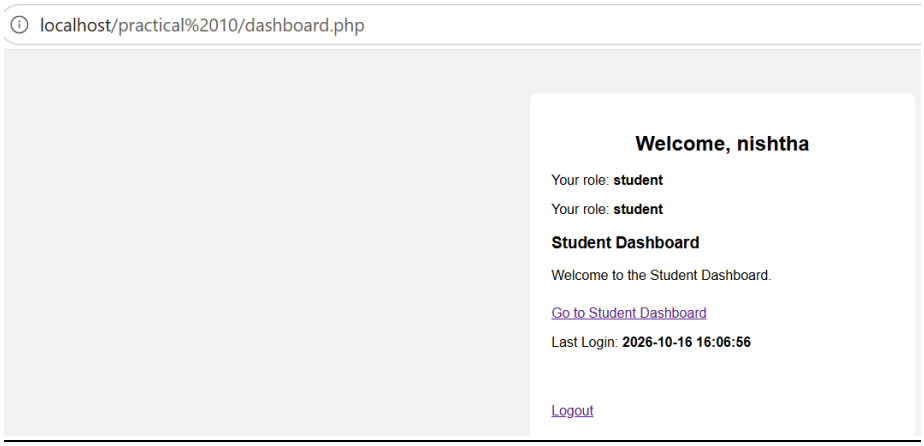
Dashboard after logout:



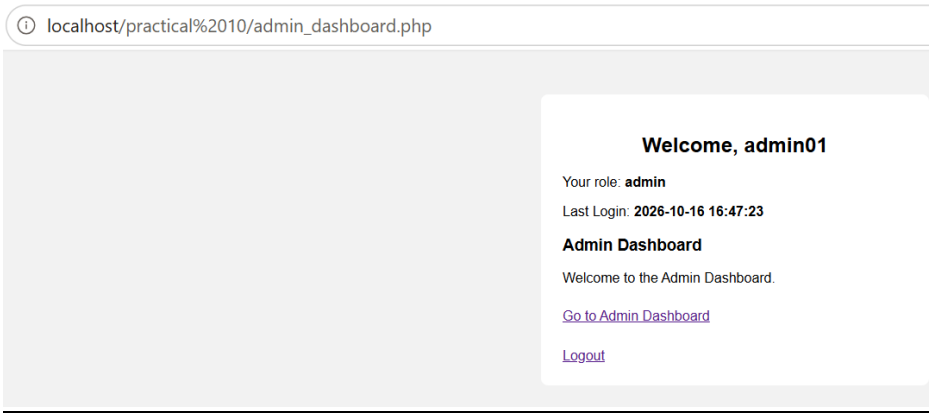
Session timeout:



student last login time:



Admin last login time:



Code:

db.php:

```php
<?php
$host = "localhost";
$username = "root";
$password = "";
$database = "user_registration";
$conn = new mysqli($host, $username, $password, $database);
if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}
$conn->set_charset("utf8mb4");
?>
```

Login.php

```php
<?php
session_start();
if (isset($_SESSION["user_id"])) {
    header("Location: dashboard.php");
    exit;
}
```

```
?>
<!DOCTYPE html>
<html>
<head>
    <title>Secure Login</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h2>Secure Login</h2>
    <?php
    if (isset($_GET["error"])) {
        echo "<p class='error'>" . htmlspecialchars($_GET["error"]) . "</p>";
    }
    if (isset($_GET["message"])) {
        echo "<p class='success'>" . htmlspecialchars($_GET["message"]) . "</p>";
    }
    ?>
    <form action="authenticate.php" method="POST">
        <label>Email:</label>
        <input type="email" name="email" required>
        <label>Password:</label>
        <input type="password" name="password" required>
        <button type="submit">Login</button>
    </form>
</div>
</body>
</html>
```

authentication.php:

```
<?php
session_start();
require_once "db.php";
if ($_SERVER["REQUEST_METHOD"] !== "POST") {
    header("Location: login.php");
    exit;
}
$email = trim($_POST["email"] ?? "");
$password = $_POST["password"] ?? "";

if ($email === "" || $password === "") {
    header("Location: login.php?error=All fields are required.");
    exit;
}

if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    header("Location: login.php?error=Invalid email address.");
    exit;
}
```

```php
$stmt = $conn->prepare( "SELECT id, username, email, password, role FROM users WHERE
email = ?");
$stmt->bind_param("s", $email);
$stmt->execute();
$result = $stmt->get_result();
if ($result->num_rows !== 1)
{
    $stmt->close();
    $conn->close();
    header("Location: login.php?error=Invalid email or password.");
    exit;
}
$user = $result->fetch_assoc();
if (!password_verify($password, $user["password"]))
{
    $stmt->close();
    $conn->close();
    header("Location: login.php?error=Invalid email or password.");
    exit;
}
session_regenerate_id(true);

$_SESSION["user_id"] = $user["id"];

$_SESSION["username"] = $user["username"];

$_SESSION["email"] = $user["email"];

$_SESSION["role"] = $user["role"];

$_SESSION["last_activity"] = time();
$stmt->close();
$conn->close();
header("Location: dashboard.php");
exit;
?>
```

Dashboard.php:

```php
<?php
session_start();
$timeout = 300;
if (!isset($_SESSION["user_id"])) {
    header("Location: login.php?error=Please login first.");
    exit;
}
if (
    isset($_SESSION["last_activity"]) && (time() - $_SESSION["last_activity"] >
$timeout)
){
    session_unset();
    session_destroy();
```

```
    header("Location: login.php?error=Session expired. Please login again.");
    exit;
}
$_SESSION["last_activity"] = time();
?>
<!DOCTYPE html>
<html>
<head>
    <title>User Dashboard</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h2>Welcome, <?php echo htmlspecialchars($_SESSION["username"]); ?></h2>
    <p> Your role:
      <strong>
            <?php echo htmlspecialchars($_SESSION["role"]); ?>
        </strong>
    </p>
    <?php if ($_SESSION["role"] === "admin"): ?>
        <h3>Admin Dashboard</h3>
        <p>Welcome to the Admin Dashboard.</p>
        <a href="admin_dashboard.php">Go to Admin Dashboard</a>
    <?php elseif ($_SESSION["role"] === "student"): ?>
        <h3>Student Dashboard</h3>
        <p>Welcome to the Student Dashboard.</p>
        <a href="student_dashboard.php">Go to Student Dashboard</a>
    <?php endif; ?>
    <br><br>
    <a href="logout.php">Logout</a>
</div>
</body>
</html>
```

admin_dashboard.php

```
<?php
session_start();
if (!isset($_SESSION["user_id"])) {
    header("Location: login.php?error=Please login first.");
    exit;
}
if ($_SESSION["role"] !== "admin") {
    http_response_code(403);
    die("Unauthorized Access: Admins only.");
}
?>
<!DOCTYPE html>
<html>
<head>
    <title>Admin Dashboard</title>
</head>
<body>
<h2>Admin Dashboard</h2>
```

```
<p>Welcome, Admin!</p>
<p>You have access to administrator features.</p>
<a href="dashboard.php">Back to Dashboard</a>
<br><br>
<a href="logout.php">Logout</a>
</body>
</html>
```

student_dashboard.php

```
<?php
session_start();
if (!isset($_SESSION["user_id"])) {
    header("Location: login.php?error=Please login first.");
    exit;
}
if ($_SESSION["role"] !== "student") {
    http_response_code(403);
    die("Unauthorized Access: Students only.");
}
?>
<!DOCTYPE html>
<html>
<head>
    <title>Student Dashboard</title>
</head>
<body>
<h2>Student Dashboard</h2>
<p>Welcome, Student!</p>
<p>You have access to student features.</p>
<a href="dashboard.php">Back to Dashboard</a>
<br><br>
<a href="logout.php">Logout</a>
</body>
</html>
```

logout.php

```
<?php
session_start();
$_SESSION = [];
if (ini_get("session.use_cookies")) {
    $params = session_get_cookie_params();
    setcookie(
        session_name(), '',time() - 42000,
        $params["path"],
        $params["domain"],
        $params["secure"],
        $params["httponly"]
    );
}
session_destroy();
header("Location: login.php?message=You have been logged out successfully.");
exit;
```

```
?>
```

timeout.php

```
<?php

session_start();

// Session timeout: 5 minutes
$timeout = 300;

if (!isset($_SESSION["user_id"])) {
    header("Location: login.php?error=Please login first.");
    exit;
}

if (
    isset($_SESSION["last_activity"]) &&
    (time() - $_SESSION["last_activity"] > $timeout)
) {
    session_unset();
    session_destroy();

    header("Location: login.php?error=Session expired. Please login again.");
    exit;
}

// Update last activity time
$_SESSION["last_activity"] = time();

?>
```

style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}
.container {
    width: 400px;
    margin: 50px auto;
    background-color: white;
    padding: 25px;
    border-radius: 8px;
}
h2 {
    text-align: center;
}
label {
    display: block;
    margin-top: 12px;
}
input {
    width: 100%;
    padding: 10px;
```

```
    margin-top: 5px;
    box-sizing: border-box;
}
button {
    width: 100%;
    padding: 10px;
    margin-top: 20px;
}
a {
    display: inline-block;
    margin-top: 10px;
}
.error {
    color: red;
}
.success {
    color: green;
}
```

User_registration.php

Table structure for table users

Column	Type	Null	Default
id	int(11)	No	
username	varchar(50)	No	
email	varchar(100)	No	
password	varchar(255)	No	
created_at	timestamp	No	current_timestamp()
role	varchar(20)	No	student

Dumping data for table users

1	nishtha	nishthap107@gmail.com	$2y$10$69gZ1OPYUGSOEGmomJUqpuH6ERZ1BJB.B0bTezfYD28IqmCLbChpm	2026-10-06 15:00:11	student
2	anushka	anushka@gmail.com	$2y$10$JGZI.qb2Sh0NAsECbqVlZu/6imrDldvfpf/6HXte3LqEUubuRk56G	2026-10-06 15:23:36	student
3	admin01	admin01@gmail.com	$2y$10$dUuPnVH2BJTUaaXdWQsA/OpzIhITcMkZMzEfjtXdlSlkCI9AM2F0O	2026-10-06 16:05:13	admin
4	student01	student01@gmail.com	$2y$10$Pjo4s1czY7WEs/1mqWfXrOxdnNKNERJAl1ZB3LdGKTPKekJwDPkze	2026-10-06 16:09:17	student

Learning outcome:

To understand and implement secure login and logout functionality using PHP sessions, verify passwords using password_verify(), regenerate session IDs after successful login, restrict unauthorized

dashboard access, implement role-based redirection for admin and student users, manage session timeouts, and understand the use of secure Remember Me tokens for maintaining user authentication.